

Brutal Series B25 - Motion & Time / Winds, Storms & Cyclones / Large Numbers / Symmetry

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A car covers the first 150 km of a trip in 2 hours and the next 90 km in 1 hour. What is its average speed for the whole journey? [\[Motion & Time\]](#)
 - (A) 240 km/h
 - (B) 120 km/h
 - (C) 82.5 km/h
 - (D) 90 km/h
 - (E) 80 km/h
 - (F) 70 km/h

2. The number 7,05,42,318 is written in the Indian system. Written in the international system, how many whole millions does it contain? [\[Large Numbers\]](#)
 - (A) 7 millions
 - (B) 7,05,42 millions
 - (C) 42 millions
 - (D) 705 millions
 - (E) 72 millions
 - (F) 700 millions
 - (G) 70 millions

3. An anemometer shows that wind carries a marker 360 m in 30 seconds. What is the wind speed in km/h? [\[Winds, Storms & Cyclones\]](#)
 - (A) 30 km/h
 - (B) 36 km/h
 - (C) 10.8 km/h
 - (D) 43.2 km/h
 - (E) 12 km/h
 - (F) 60 km/h

4. A regular hexagon is given. State its number of lines of symmetry, its order of rotational symmetry, and the smallest angle of rotation that maps it onto itself. [\[Symmetry\]](#)
 - (A) 5 lines; order 5; 72 degrees
 - (B) 6 lines; order 6; 90 degrees
 - (C) 6 lines; order 6; 60 degrees
 - (D) 12 lines; order 12; 30 degrees
 - (E) 3 lines; order 3; 120 degrees
 - (F) 6 lines; order 3; 60 degrees

5. A pendulum completes 30 oscillations in 60 seconds. Keeping the same swing, how many oscillations will it complete in 5 minutes? [\[Motion & Time\]](#)
- (A) 25
 - (B) 60
 - (C) 150
 - (D) 120
 - (E) 75
 - (F) 300
6. Round 4,87,650 and 2,13,420 each to the nearest thousand, then add the rounded values. What total do you get? [\[Large Numbers\]](#)
- (A) 6,99,000
 - (B) 7,01,070
 - (C) 7,02,000
 - (D) 7,00,000
 - (E) 7,01,000
 - (F) 6,89,000
 - (G) 7,00,500
7. Place A has an air pressure of 1018 units and place B has 1004 units. In which direction does the wind blow, and what is the pressure difference? [\[Winds, Storms & Cyclones\]](#)
- (A) From A to B; difference 14 units
 - (B) From B to A; difference 7 units
 - (C) From B to A; difference 1004 units
 - (D) From A to B; difference 2022 units
 - (E) No wind; difference 0
 - (F) From A to B; difference 22 units
 - (G) From B to A; difference 14 units
8. Consider the capital letter H. How many lines of symmetry does it have, and what is its order of rotational symmetry? [\[Symmetry\]](#)
- (A) 4 lines; rotational order 4
 - (B) 0 lines; rotational order 2
 - (C) 1 line; rotational order 2
 - (D) 2 lines; rotational order 2
 - (E) 2 lines; rotational order 4
 - (F) 2 lines; rotational order 1
 - (G) 1 line; rotational order 1
9. A motorbike travels at 72 km/h. Converting this to metres per second, how far does it go in 15 seconds? [\[Motion & Time\]](#)
- (A) 72 m
 - (B) 75 m
 - (C) 300 m
 - (D) 200 m
 - (E) 1080 m
 - (F) 20 m
 - (G) 450 m

- 10.** In the number 58,46,392, subtract the place value of the digit 4 from the place value of the digit 8. What is the result? [\[Large Numbers\]](#)
- (A) 40,000
 - (B) 84,000
 - (C) 4
 - (D) 7,60,000
 - (E) 7,99,996
 - (F) 8,40,000
- 11.** On a hot afternoon at the seaside, you feel a cool breeze coming from the sea toward the land. What causes this sea breeze? [\[Winds, Storms & Cyclones\]](#)
- (A) Land heats faster, air over it rises, and cooler sea air moves in
 - (B) Land cools faster than the sea by day
 - (C) The sea is always windier than land
 - (D) Salt in the sea pushes air to land
 - (E) Sea heats faster, so air rushes out to sea
 - (F) The Moon pulls air toward the shore
- 12.** An equilateral triangle is rotated about its centre. Through what smallest angle must it turn to look exactly the same, and what is its order of rotational symmetry? [\[Symmetry\]](#)
- (A) 180 degrees; order 3
 - (B) 60 degrees; order 6
 - (C) 360 degrees; order 1
 - (D) 120 degrees; order 3
 - (E) 120 degrees; order 4
 - (F) 120 degrees; order 2
 - (G) 90 degrees; order 4
- 13.** A car's odometer reads 25,430 km at the start and 25,710 km after 3.5 hours of driving. What is its average speed? [\[Motion & Time\]](#)
- (A) 70 km/h
 - (B) 90 km/h
 - (C) 8 km/h
 - (D) 280 km/h
 - (E) 800 km/h
 - (F) 51,140 km/h
 - (G) 80 km/h
- 14.** What is the predecessor of one crore (1,00,00,000)? [\[Large Numbers\]](#)
- (A) 99,99,990
 - (B) 10,00,000
 - (C) 1,00,00,000
 - (D) 99,99,999
 - (E) 9,99,999
 - (F) 99,99,998
 - (G) 1,00,00,001

15. When air is heated, how do its volume, density and movement change? [\[Winds, Storms & Cyclones\]](#)

- (A) Volume decreases, density decreases, and it rises
- (B) Volume stays same, density decreases, and it sinks
- (C) Volume increases, density increases, and it rises
- (D) Nothing changes when air is heated
- (E) Volume increases, density decreases, and it rises
- (F) Volume decreases, density increases, and it sinks
- (G) Volume increases, density decreases, and it sinks

16. A square is given. How many lines of symmetry does it have, and what is the smallest angle of rotation that maps it onto itself? [\[Symmetry\]](#)

- (A) 4 lines; 90 degrees
- (B) 2 lines; 90 degrees
- (C) 2 lines; 180 degrees
- (D) 4 lines; 360 degrees
- (E) 4 lines; 45 degrees
- (F) 4 lines; 60 degrees
- (G) 8 lines; 45 degrees

17. A train 120 m long passes a standing pole in 6 seconds. What is the train's speed in km/h? [\[Motion & Time\]](#)

- (A) 72 km/h
- (B) 60 km/h
- (C) 36 km/h
- (D) 20 km/h
- (E) 720 km/h
- (F) 12 km/h
- (G) 100 km/h

18. A factory makes about 9,850 units a day. Estimate the yearly output by rounding the daily figure to the nearest thousand and using 365 days, then find how far this estimate is above the exact total. [\[Large Numbers\]](#)

- (A) Estimate 39,40,000; 54,750 above exact
- (B) Estimate 36,50,000; 1,50,000 above exact
- (C) Estimate 36,50,000; 5,475 above exact
- (D) Estimate 35,00,000; 95,250 above exact
- (E) Estimate 98,50,000; 54,750 above exact
- (F) Estimate 36,50,000; about 54,750 above exact
- (G) Estimate 35,95,250; 0 above exact

19. A cyclone's centre moves 270 km in 6 hours. At this rate, what is the speed of the cyclone's movement? [\[Winds, Storms & Cyclones\]](#)

- (A) 270 km/h
- (B) 90 km/h
- (C) 54 km/h
- (D) 6 km/h
- (E) 45 km/h
- (F) 40 km/h
- (G) 1620 km/h

- 20.** A regular pentagon has rotational symmetry. What is the smallest angle through which it can be turned to look the same? [\[Symmetry\]](#)
- (A) 45 degrees
 - (B) 72 degrees
 - (C) 120 degrees
 - (D) 108 degrees
 - (E) 90 degrees
 - (F) 36 degrees
- 21.** Two cars leave the same spot at the same moment in opposite directions at 40 km/h and 60 km/h. How far apart are they after 2.5 hours? [\[Motion & Time\]](#)
- (A) 500 km
 - (B) 125 km
 - (C) 50 km
 - (D) 200 km
 - (E) 150 km
 - (F) 250 km
- 22.** Read the number 3,05,06,007 in the Indian system. Which words name it correctly? [\[Large Numbers\]](#)
- (A) Three crore five lakh six thousand seventy
 - (B) Three crore five lakh six thousand seven
 - (C) Three crore five lakh sixty thousand seven
 - (D) Thirty crore five lakh six thousand seven
 - (E) Three crore fifty lakh six thousand seven
 - (F) Thirty lakh fifty-six thousand seven
- 23.** Holding a thin paper strip just below your lips and blowing hard across its top makes the loose end swing up. What explains this? [\[Winds, Storms & Cyclones\]](#)
- (A) Gravity weakens when air moves
 - (B) Fast-moving air has lower pressure, so higher pressure below pushes the paper up
 - (C) Sound waves push the paper up
 - (D) Your breath is warm and lifts the paper
 - (E) Static charge lifts the paper
 - (F) The paper becomes lighter when blown on
 - (G) Fast air has higher pressure and pushes up
- 24.** A rectangle that is not a square is given. How many lines of symmetry does it have, and what is its order of rotational symmetry? [\[Symmetry\]](#)
- (A) 0 lines; order 2
 - (B) 1 line; order 2
 - (C) 2 lines; order 4
 - (D) 2 lines; order 2
 - (E) 4 lines; order 2
 - (F) 4 lines; order 4
 - (G) 2 lines; order 1

25. An object moves with uniform speed, covering 12 m every 4 seconds. How far does it travel in 20 seconds? [\[Motion & Time\]](#)

- (A) 3 m
- (B) 80 m
- (C) 60 m
- (D) 240 m
- (E) 48 m
- (F) 12 m

26. A city tank holds 12,50,000 litres. After 8,75,400 litres are used, how much water remains? [\[Large Numbers\]](#)

- (A) 3,74,600 litres
- (B) 3,74,400 litres
- (C) 2,74,600 litres
- (D) 21,25,400 litres
- (E) 4,74,600 litres
- (F) 3,75,400 litres
- (G) 4,25,400 litres

27. Why do winds blow at all, in terms of air pressure? [\[Winds, Storms & Cyclones\]](#)

- (A) Winds need no pressure difference
- (B) Air stays still unless pushed by rain
- (C) Winds blow from light areas to dark areas
- (D) Air moves only where it is coldest
- (E) Air moves from low temperature to high temperature
- (F) Air moves from low pressure to high pressure
- (G) Air moves from regions of high pressure to regions of low pressure

28. A perfect circle is drawn on paper. State the number of lines of symmetry it has together with its order of rotational symmetry. [\[Symmetry\]](#)

- (A) No lines; order 1
- (B) Infinitely many lines; order 2
- (C) 2 lines; infinite order
- (D) 1 line; order 1
- (E) 4 lines; order 4
- (F) 360 lines; order 360
- (G) Infinitely many lines; infinite order

29. A pendulum clock's bob has a time period of 2 seconds. How many oscillations does it make in one full day? [\[Motion & Time\]](#)

- (A) 1,440
- (B) 43,200
- (C) 2,400
- (D) 720
- (E) 86,400
- (F) 1,72,800
- (G) 21,600

30. Which of these is the largest number: 9,08,76,543 or 9,80,67,345 or 9,08,67,543? [\[Large Numbers\]](#)

- (A) All are equal
- (B) 9,08,76,543
- (C) 9,80,67,543 and 9,08,76,543 tie
- (D) 9,80,67,345
- (E) Cannot be decided
- (F) 9,08,67,543

31. During the Indian summer, the south-west monsoon brings heavy rain. Where does this moist wind mainly come from? [\[Winds, Storms & Cyclones\]](#)

- (A) From the poles toward the equator with no moisture
- (B) From space toward Earth
- (C) From the sea toward the land
- (D) From the land toward the sea
- (E) From the cold mountains down to the plains
- (F) From the dry desert only
- (G) From the land toward the mountains only

32. The capital letter S has how many lines of symmetry, and what is its order of rotational symmetry?

[\[Symmetry\]](#)

- (A) 1 line; rotational order 1
- (B) 0 lines; rotational order 1
- (C) 2 lines; rotational order 2
- (D) 1 line; rotational order 2
- (E) 0 lines; rotational order 4
- (F) 2 lines; rotational order 1
- (G) 0 lines; rotational order 2

33. A traveller covers 60 km at 30 km/h and then another 60 km at 60 km/h. What is the average speed for the full 120 km? [\[Motion & Time\]](#)

- (A) 45 km/h
- (B) 90 km/h
- (C) 50 km/h
- (D) 40 km/h
- (E) 30 km/h
- (F) 48 km/h
- (G) 60 km/h

34. Round 6,48,500 to the nearest lakh. What do you get? [\[Large Numbers\]](#)

- (A) 6,00,000
- (B) 5,00,000
- (C) 6,40,000
- (D) 6,48,000
- (E) 6,50,000
- (F) 6,49,000
- (G) 7,00,000

35. A cyclone's strongest winds blow at 150 km/h. How far could such a wind carry loose debris in 20 minutes? [\[Winds, Storms & Cyclones\]](#)

- (A) 50 km
- (B) 450 km
- (C) 30 km
- (D) 100 km
- (E) 150 km
- (F) 75 km
- (G) 3000 km

36. Which capital letter below has BOTH a vertical and a horizontal line of symmetry? [\[Symmetry\]](#)

- (A) X
- (B) A
- (C) R
- (D) E
- (E) B
- (F) P

37. A cheetah runs at 25 m/s. How many kilometres does it cover if it keeps this up for 4 minutes? [\[Motion & Time\]](#)

- (A) 90 km
- (B) 100 km
- (C) 360 km
- (D) 0.6 km
- (E) 6 km
- (F) 60 km

38. One crore can be written as a power of ten. How many zeros does 1 crore have, and what power of ten is it? [\[Large Numbers\]](#)

- (A) 5 zeros; 10 to the power 5
- (B) 8 zeros; 10 to the power 8
- (C) 6 zeros; 10 to the power 6
- (D) 9 zeros; 10 to the power 9
- (E) 7 zeros; 10 to the power 7
- (F) 7 zeros; 10 to the power 6

39. A barometer in a coastal town shows the air pressure falling quickly over a few hours. What does this most likely mean? [\[Winds, Storms & Cyclones\]](#)

- (A) Clear, calm weather is settling in
- (B) Rainfall has just stopped for good
- (C) The town has moved to higher ground
- (D) An earthquake is coming
- (E) A storm or cyclone may be approaching
- (F) The Sun is about to set
- (G) The temperature is about to drop to freezing

- 40.** A shape looks exactly the same after every rotation of 45 degrees about its centre. What is its order of rotational symmetry? [\[Symmetry\]](#)
- (A) 8
 - (B) 45
 - (C) 6
 - (D) 16
 - (E) 9
 - (F) 2
- 41.** A runner completes a 400 m track in 80 seconds. At the same speed, how long will he take to run 1 km? [\[Motion & Time\]](#)
- (A) 3 minutes 20 seconds
 - (B) 1 minute 40 seconds
 - (C) 100 seconds
 - (D) 5 minutes
 - (E) 320 seconds
 - (F) 4 minutes
- 42.** Using each of the digits 3, 0, 5, 7 exactly once, find the difference between the greatest and the smallest four-digit numbers you can make. [\[Large Numbers\]](#)
- (A) 4,752
 - (B) 4,347
 - (C) 4,257
 - (D) 7,047
 - (E) 7,530
 - (F) 4,473
- 43.** In the classic experiment, a tin can with a little water is heated until steam pours out, then sealed and cooled. The can gets crushed. Why? [\[Winds, Storms & Cyclones\]](#)
- (A) The can rusts and weakens suddenly
 - (B) Cooling lowers the pressure inside, so the higher outside air pressure crushes it
 - (C) The water freezes and expands inside
 - (D) The can shrinks because metal always shrinks a lot
 - (E) Steam expands and bursts the can outward
 - (F) Gravity increases as the can cools
 - (G) The heat melts the metal of the can
- 44.** A regular polygon has rotational symmetry of order 5. What is its smallest angle of rotation, and how many lines of symmetry does it have? [\[Symmetry\]](#)
- (A) 36 degrees; 5 lines
 - (B) 108 degrees; 5 lines
 - (C) 90 degrees; 5 lines
 - (D) 72 degrees; 0 lines
 - (E) 60 degrees; 6 lines
 - (F) 72 degrees; 5 lines

45. A cyclist rides at 18 km/h. Converting to m/s, how many seconds does she take to cross a 150 m bridge? [\[Motion & Time\]](#)

- (A) 8.3 seconds
- (B) 60 seconds
- (C) 25 seconds
- (D) 150 seconds
- (E) 5 seconds
- (F) 30 seconds
- (G) 45 seconds

46. Estimate the product 612×488 by rounding each number to the nearest hundred, then state how this estimate compares with the exact product. [\[Large Numbers\]](#)

- (A) Estimate 3,00,000; it is higher than the exact product
- (B) Estimate 2,40,000; it is higher than the exact product
- (C) Estimate 3,00,000; it equals the exact product
- (D) Estimate 1,10,000; it is higher
- (E) Estimate 2,98,656; it is the exact product
- (F) Estimate 3,60,000; it is lower than the exact product

47. A cyclone's centre advances steadily at 36 km/h toward a town that is 90 km away. How long before it reaches the town? [\[Winds, Storms & Cyclones\]](#)

- (A) 4 hours
- (B) 90 hours
- (C) 1.5 hours
- (D) 2 hours
- (E) 2.5 hours
- (F) 3.24 hours

48. A parallelogram that is not a rectangle or rhombus is given. How many lines of symmetry does it have, and what is its order of rotational symmetry? [\[Symmetry\]](#)

- (A) 2 lines; order 2
- (B) 2 lines; order 1
- (C) 1 line; order 1
- (D) 0 lines; order 4
- (E) 0 lines; order 2
- (F) 4 lines; order 4
- (G) 0 lines; order 1

49. Pendulum A makes 20 oscillations in 40 seconds; pendulum B makes 30 oscillations in 45 seconds. Which swings faster, and how many more oscillations does the faster one make in 90 seconds? [\[Motion & Time\]](#)

- (A) A; 15 more
- (B) A; 45 more
- (C) B; 5 more
- (D) B; 15 more
- (E) A; 10 more
- (F) Both equal; 0 more
- (G) B; 30 more

50. Three towns have populations 4,56,789, 3,08,910 and 2,34,301. What is their combined population?

[Large Numbers]

- (A) 9,90,000
- (B) 10,00,100
- (C) 9,00,000
- (D) 10,00,000
- (E) 11,00,000
- (F) 10,10,000
- (G) 9,99,000

51. Over an hour the wind speed at a station rises from 20 km/h to 25 km/h. By what percentage did the wind speed increase? [Winds, Storms & Cyclones]

- (A) 50%
- (B) 45%
- (C) 25%
- (D) 5%
- (E) 125%
- (F) 80%
- (G) 20%

52. A regular octagon is given. How many lines of symmetry does it have, and what is its smallest angle of rotational symmetry? [Symmetry]

- (A) 8 lines; 22.5 degrees
- (B) 8 lines; 60 degrees
- (C) 4 lines; 45 degrees
- (D) 8 lines; 90 degrees
- (E) 8 lines; 45 degrees
- (F) 6 lines; 60 degrees
- (G) 16 lines; 45 degrees

53. A bus runs at 60 km/h for 30 minutes, then at 80 km/h for 45 minutes. What total distance does it cover? [Motion & Time]

- (A) 140 km
- (B) 100 km
- (C) 70 km
- (D) 105 km
- (E) 120 km
- (F) 90 km
- (G) 72 km

54. What is the place value of the digit 6 in the number 9,67,40,285? [Large Numbers]

- (A) 6,000
- (B) 6
- (C) 60,00,000
- (D) 6,40,285
- (E) 6,00,00,000
- (F) 60,000
- (G) 6,00,000

55. Why does air over the equator tend to rise while air over the poles tends to sink, helping to set up large-scale winds? [\[Winds, Storms & Cyclones\]](#)

- (A) The poles are heated more, so polar air rises
- (B) The equator is heated more, so its warm air rises, while cold polar air sinks
- (C) Air rises wherever it is coldest
- (D) The equator is colder than the poles
- (E) Winds need no temperature difference
- (F) Polar air is lighter than equatorial air

56. A figure first looks the same when turned by 60 degrees about its centre. What is its order of rotational symmetry, and what other turn (under a full circle) also maps it onto itself? [\[Symmetry\]](#)

- (A) Order 60; only 60 degrees
- (B) Order 3; also 120, 240 degrees
- (C) Order 6; only 360 degrees
- (D) Order 12; also 30, 60 degrees
- (E) Order 6; also 120, 180, 240, 300 degrees
- (F) Order 2; also 180 degrees

57. On a distance-time graph, an object's line is horizontal (flat) for 10 seconds. What does this tell us about the object during that time? [\[Motion & Time\]](#)

- (A) It is at rest (not moving)
- (B) It moves at 10 m/s
- (C) It slows down to a stop
- (D) It speeds up
- (E) It moves at a steady high speed
- (F) It moves backward

58. Convert 5 million into the Indian system. How many lakhs is that? [\[Large Numbers\]](#)

- (A) 5,000 lakhs
- (B) 50 lakhs
- (C) 500 lakhs
- (D) 0.5 lakh
- (E) 5 crores
- (F) 50,000 lakhs

59. At night by the coast, a breeze blows from the land out toward the sea. What is the main reason for this land breeze? [\[Winds, Storms & Cyclones\]](#)

- (A) Land cools faster than the sea, so cooler land air flows toward the warmer sea
- (B) The sea cools faster than the land at night
- (C) The Moon heats the sea at night
- (D) Winds always blow toward the sea at night for no reason
- (E) Salt water pulls air toward the shore
- (F) Land stays hotter than the sea all night
- (G) Stars warm the land at night

60. A scalene triangle (all sides different) is given. How many lines of symmetry does it have? [\[Symmetry\]](#)

- (A) 2
- (B) 0
- (C) 4
- (D) Infinitely many
- (E) 3
- (F) 6
- (G) 1

61. A train covers 90 km in 1 hour 15 minutes. What is its speed in km/h? [\[Motion & Time\]](#)

- (A) 72 km/h
- (B) 90 km/h
- (C) 75 km/h
- (D) 112.5 km/h
- (E) 6 km/h
- (F) 68 km/h
- (G) 60 km/h

62. A number rounded to the nearest thousand is 47,000, and rounded to the nearest hundred is 46,800. Which of these could the exact number be? [\[Large Numbers\]](#)

- (A) 47,489
- (B) 46,749
- (C) 46,789
- (D) 46,851
- (E) 46,749 and 47,489 both
- (F) 46,500
- (G) 47,051

63. An anemometer's cups complete 90 full rotations in 60 seconds. How many rotations is that per second? [\[Winds, Storms & Cyclones\]](#)

- (A) 0.67 rotations per second
- (B) 60 rotations per second
- (C) 90 rotations per second
- (D) 1.5 rotations per second
- (E) 30 rotations per second
- (F) 2 rotations per second
- (G) 15 rotations per second

64. An isosceles triangle (exactly two equal sides) is given. How many lines of symmetry does it have, and what is its order of rotational symmetry? [\[Symmetry\]](#)

- (A) 2 lines; order 1
- (B) 2 lines; order 2
- (C) 0 lines; order 1
- (D) 1 line; order 3
- (E) 1 line; order 2
- (F) 3 lines; order 3
- (G) 1 line; order 1

65. A bus moves at 5 m/s. Converting to km/h first, how long does it take to travel 9 km? [\[Motion & Time\]](#)

- (A) 15 minutes
- (B) 1 hour
- (C) 1800 seconds only if not converted
- (D) 18 minutes
- (E) 45 minutes
- (F) 30 minutes
- (G) 9 minutes

66. Write the expanded form value: in 80,04,506, what is the sum of the place values of the two non-zero digits in the thousands group and hundreds place? [\[Large Numbers\]](#)

- (A) 4,506
- (B) 4,560
- (C) 4,500
- (D) 45,060
- (E) 4,006
- (F) 4,000

67. Strong wind blows fast over a flat roof while the air inside the house is still. What is the danger, and why? [\[Winds, Storms & Cyclones\]](#)

- (A) The still air inside has the lowest pressure
- (B) The roof is pulled down hard by the fast wind
- (C) Fast air above has the highest pressure, sealing the roof down
- (D) Nothing happens, since wind only pushes sideways
- (E) The roof can be pushed up and off, because fast air above has lower pressure
- (F) The roof gets heavier and cracks
- (G) The roof cools and shrinks away

68. A rhombus that is not a square is given. How many lines of symmetry does it have, and where do they lie? [\[Symmetry\]](#)

- (A) 2 lines; along its two diagonals
- (B) 1 line; along one diagonal
- (C) 0 lines; it has none
- (D) 2 lines; through the midpoints of opposite sides
- (E) 4 lines; along the diagonals
- (F) 4 lines; through all sides and diagonals

69. A car's average speed over a 4-hour trip is found from an odometer rise of 360 km. At the same average speed, how far will it go in the next 2 hours? [\[Motion & Time\]](#)

- (A) 270 km
- (B) 160 km
- (C) 90 km
- (D) 180 km
- (E) 360 km
- (F) 720 km
- (G) 45 km

70. Arrange these in ascending order and pick the second-smallest: 1,23,456 ; 1,32,456 ; 1,23,654 ; 1,22,999. [\[Large Numbers\]](#)

- (A) 1,22,456
- (B) 1,23,456 and 1,23,654 tie
- (C) 1,23,654
- (D) 1,33,000
- (E) 1,23,456
- (F) 1,22,999

71. A coastal district issues a cyclone warning when the storm is 200 km away and moving toward it at 40 km/h. How many hours of warning does this give before landfall? [\[Winds, Storms & Cyclones\]](#)

- (A) 40 hours
- (B) 2.5 hours
- (C) 200 hours
- (D) 5 hours
- (E) 8000 hours
- (F) 4 hours

72. A regular 12-sided polygon (regular dodecagon) is given. What is its smallest angle of rotational symmetry, and how many lines of symmetry does it have? [\[Symmetry\]](#)

- (A) 15 degrees; 12 lines
- (B) 30 degrees; 24 lines
- (C) 60 degrees; 12 lines
- (D) 30 degrees; 6 lines
- (E) 12 degrees; 12 lines
- (F) 36 degrees; 10 lines
- (G) 30 degrees; 12 lines

73. A pendulum makes 25 oscillations in 50 seconds. If the string is made shorter, what happens to its time period and how fast it swings? [\[Motion & Time\]](#)

- (A) Oscillations stop completely
- (B) Time period gets larger; it swings slower
- (C) Time period gets smaller; it swings faster
- (D) Time period gets larger; it swings faster
- (E) Time period becomes zero; it stops
- (F) Time period stays 2 s; no change
- (G) Time period gets smaller; it swings slower

74. The face value of a digit minus its place value can be huge. In 7,40,000 what is the place value of 7 minus its face value? [\[Large Numbers\]](#)

- (A) 6,93,000
- (B) 6,99,000
- (C) 7,39,993
- (D) 7,00,000
- (E) 6,99,993
- (F) 6,99,993 is wrong, it is 0
- (G) 7,40,000

- 75.** Which combination correctly describes the calm centre of a cyclone and the air around it? [[Winds, Storms & Cyclones](#)]
- (A) Calm air everywhere inside the cyclone
 - (B) A calm 'eye' at the centre, surrounded by very strong winds and heavy rain
 - (C) A stormy centre surrounded by calm clear air
 - (D) Strong winds at the centre, calm at the edges only
 - (E) A frozen centre surrounded by warm winds
 - (F) No winds anywhere in a cyclone
- 76.** The capital letter N has how many lines of symmetry, and what is its order of rotational symmetry? [[Symmetry](#)]
- (A) 1 line; rotational order 1
 - (B) 2 lines; rotational order 1
 - (C) 0 lines; rotational order 2
 - (D) 0 lines; rotational order 1
 - (E) 0 lines; rotational order 4
 - (F) 1 line; rotational order 2
 - (G) 2 lines; rotational order 2
- 77.** A car covers the first 100 km at 50 km/h and the next 100 km at 25 km/h. What is its average speed for the 200 km? [[Motion & Time](#)]
- (A) 50 km/h
 - (B) 75 km/h
 - (C) 30 km/h
 - (D) 40 km/h
 - (E) About 33.3 km/h
 - (F) 37.5 km/h
 - (G) 25 km/h
- 78.** How many seconds are there in one full day, written as a single large number? [[Large Numbers](#)]
- (A) 86,400
 - (B) 1,440
 - (C) 43,200
 - (D) 24,000
 - (E) 86,000
 - (F) 8,64,000
 - (G) 3,600
- 79.** Warm, moist air rising rapidly is the starting point of a thunderstorm. What two things does this rising air do as it goes higher and cools? [[Winds, Storms & Cyclones](#)]
- (A) It freezes the ground below
 - (B) It warms further and dries out completely
 - (C) It stops moving and disappears
 - (D) It turns into solid air
 - (E) It cools but releases no water at all
 - (F) It cools and its water vapour condenses into clouds and rain

80. A regular polygon has each smallest rotation of 40 degrees mapping it onto itself. How many sides does it have, and how many lines of symmetry? [\[Symmetry\]](#)

- (A) 8 sides; 8 lines
- (B) 9 sides; 9 lines
- (C) 6 sides; 6 lines
- (D) 40 sides; 40 lines
- (E) 9 sides; 18 lines
- (F) 9 sides; 0 lines
- (G) 10 sides; 10 lines

81. An aeroplane cruises at 15 m/s during taxi. How far does it travel in 8 minutes at this speed? [\[Motion & Time\]](#)

- (A) 120 m
- (B) 7.2 km
- (C) 1.2 km
- (D) 2 km
- (E) 7200 km
- (F) 54 km
- (G) 720 m

82. A charity collected 12,07,560 rupees in one drive and 9,92,440 in the next. What is the total collected, and how is it read? [\[Large Numbers\]](#)

- (A) 22,00,000 rupees; twenty-two lakh rupees
- (B) 22,00,000 rupees; two crore two lakh
- (C) 20,00,000 rupees; twenty lakh
- (D) 22,00,000 rupees; twenty-two thousand
- (E) 22,99,000 rupees; twenty-two lakh ninety-nine thousand
- (F) 21,00,000 rupees; twenty-one lakh

83. Wind speed at a kite-flying ground jumps from 30 km/h to 48 km/h. Expressed as a fraction in lowest terms, the new speed is what multiple of the old speed? [\[Winds, Storms & Cyclones\]](#)

- (A) $\frac{3}{5}$ of the old speed
- (B) $\frac{5}{8}$ of the old speed
- (C) 2 times the old speed
- (D) $\frac{18}{30}$ of the old speed
- (E) $\frac{4}{3}$ of the old speed
- (F) $\frac{48}{30}$ unsimplified is the only answer
- (G) $\frac{8}{5}$ of the old speed

84. Which statement is TRUE for a shape that has rotational symmetry of order 4? [\[Symmetry\]](#)

- (A) It looks the same after turns of 90, 180, 270 and 360 degrees
- (B) It looks the same after turns of 60 degrees
- (C) It looks the same after turns of 120 degrees
- (D) It has exactly 4 lines of symmetry always
- (E) It looks the same only after a 360-degree turn
- (F) It can never have any line of symmetry
- (G) It looks the same after turns of 45 degrees

85. A car travels at 90 km/h. If the driver's reaction time is 2 seconds, how far does the car move before the brakes are even pressed? [\[Motion & Time\]](#)

- (A) 90 m
- (B) 50 m
- (C) 100 m
- (D) 180 m
- (E) 25 m
- (F) 2 m
- (G) 45 m

86. Estimate the quotient 5,94,000 divided by 29 by first rounding the divisor to the nearest ten. What is the estimate? [\[Large Numbers\]](#)

- (A) 19,800 is wrong, it is 297
- (B) 19,800
- (C) 30,000
- (D) 19,000
- (E) 59,400
- (F) 2,049
- (G) 20,483

87. Why does a tightly shut house still suffer when a cyclone passes, even if no window breaks? [\[Winds, Storms & Cyclones\]](#)

- (A) Air inside vanishes during a cyclone
- (B) The sudden low outside pressure can make walls and roofs bulge or lift
- (C) The house spins with the cyclone
- (D) The low pressure makes the house heavier
- (E) Shut windows raise the inside temperature dangerously only
- (F) Cyclones only affect open fields

88. A kite (the quadrilateral shape) that is not a rhombus is given. How many lines of symmetry does it have, and what is its order of rotational symmetry? [\[Symmetry\]](#)

- (A) 0 lines; order 1
- (B) 4 lines; order 4
- (C) 1 line; order 1
- (D) 2 lines; order 1
- (E) 1 line; order 2
- (F) 2 lines; order 2
- (G) 1 line; order 4

89. A pendulum completes 60 oscillations in 90 seconds. How many oscillations will it complete in 3 minutes? [\[Motion & Time\]](#)

- (A) 60
- (B) 240
- (C) 45
- (D) 120
- (E) 180
- (F) 90
- (G) 30

- 90.** In 6,30,05,070, which digit has the place value of zero contributing nothing, and what is the place value of the digit 3? [\[Large Numbers\]](#)
- (A) The 3 has place value 3,000
 - (B) The 3 has place value 3
 - (C) Several zeros contribute nothing; the 3 has place value 30,00,000
 - (D) The 3 has place value 3,00,00,000
 - (E) The 3 has place value 30,00,00,000
 - (F) The 3 has place value 3,00,000
- 91.** A weather balloon records wind speed as 15 m/s near the ground. A pilot wants this in km/h. What is the value, and is it a gentle or strong wind for the ground? [\[Winds, Storms & Cyclones\]](#)
- (A) 150 km/h; a strong wind
 - (B) 4.2 km/h; a gentle wind
 - (C) 75 km/h; a strong wind
 - (D) 54 km/h; a fairly strong wind
 - (E) 15 km/h; a gentle wind
 - (F) 5 km/h; a calm wind
- 92.** Two regular polygons are compared. Polygon X has 6 lines of symmetry and polygon Y has rotational symmetry of order 3. What is the difference between their smallest rotation angles? [\[Symmetry\]](#)
- (A) 0 degrees
 - (B) 30 degrees
 - (C) 120 degrees
 - (D) 60 degrees
 - (E) 90 degrees
 - (F) 3 degrees
 - (G) 180 degrees
- 93.** A car's odometer rises by 48 km in 40 minutes of steady driving. What is its speed in km/h? [\[Motion & Time\]](#)
- (A) 72 km/h
 - (B) 120 km/h
 - (C) 1.2 km/h
 - (D) 32 km/h
 - (E) 80 km/h
 - (F) 48 km/h
 - (G) 60 km/h
- 94.** A printing press prints 2,48,000 copies a month. Rounding to the nearest ten thousand, about how many copies does it print in a year (12 months)? [\[Large Numbers\]](#)
- (A) 2,50,000
 - (B) 29,76,000
 - (C) 36,00,000
 - (D) 3,00,000
 - (E) 24,80,000
 - (F) 30,00,000 is wrong, it is 2,50,000
 - (G) 30,00,000

- 95.** Two weather stations are 60 km apart. A storm front passes the first at 9:00 and the second at 10:30. What is the front's speed? [\[Winds, Storms & Cyclones\]](#)
- (A) 90 km/h
 - (B) 30 km/h
 - (C) 1.5 km/h
 - (D) 40 km/h
 - (E) 20 km/h
 - (F) 60 km/h
 - (G) 45 km/h
- 96.** How many of the capital letters in the word 'MATH' have at least one line of symmetry? [\[Symmetry\]](#)
- (A) M and A only
 - (B) 1
 - (C) 0
 - (D) H only
 - (E) 3
 - (F) 4
- 97.** A fast train at 54 km/h and a slow train at 36 km/h move along parallel tracks in the same direction, starting 9 km apart with the fast one behind. How long until the fast train catches up? [\[Motion & Time\]](#)
- (A) 30 minutes
 - (B) 1 hour
 - (C) 18 minutes
 - (D) 90 minutes
 - (E) 12 minutes
 - (F) 6 minutes
 - (G) 10 minutes
- 98.** A satellite circles the Earth once every 90 minutes. How many complete orbits does it make in 7 full days, written as a large number? [\[Large Numbers\]](#)
- (A) 672 orbits
 - (B) 112 orbits
 - (C) 160 orbits
 - (D) 16 orbits
 - (E) 1,008 orbits
 - (F) 11,200 orbits
- 99.** Why is it dangerous to take shelter under a lone tall tree during a thunderstorm, in terms of how storms behave? [\[Winds, Storms & Cyclones\]](#)
- (A) Tall trees make the wind much warmer
 - (B) Standing under a tree stops the wind completely and is safe
 - (C) Trees absorb all the storm's energy harmlessly
 - (D) The tall tree can attract a lightning strike, which can jump to a person beneath it
 - (E) The tree turns the storm away
 - (F) A lone tree blocks all the rain safely
 - (G) Trees lower the air pressure to safe levels

100. A design looks the same after a 90-degree turn but has no line of symmetry at all. Which description fits it? [\[Symmetry\]](#)

- (A) Rotational symmetry of order 4, but zero lines of symmetry
- (B) Rotational order 2 and no lines
- (C) Rotational order 1 and 4 lines
- (D) Rotational order 4 and 1 line
- (E) No rotational symmetry and no lines
- (F) Rotational order 4 and 4 lines of symmetry